1-3-1: Introduction to TCP_IP and the OSI Model

At the end of this episode, I will be able to:

1. Identify and explain the importance of the TCP/IP and OSI models, given a scenario.

Learner Objective: *Identify and explain the importance of the TCP/IP and OSI models. given a scenario*

Description: In this episode, the learner will examine the TCP/IP and Open Systems Interconnection Models. We will explore the conceptual layers and importance of each layer.

- Where did the OSI model come from?
  - The International Organization for Standardization (ISO), a multinational body dedicated to worldwide standardization.
- What is the OSI Model?
  - Open Systems Interconnection Model
  - A conceptual framework used for standardizing the functions of telecommunication and computing systems
- Layer Structure
  - Application Layer (L7) - handles the network services to end-user applications
  - Presentation Layer (L6) - Translates and encrypts data for the application layer
  - Session Layer (L5) - Manages sessions and dialogues between applications
  - Transport Layer (L4) - Ensures complete and reliable data transfer
  - Network Layer (L3)- Responsible for data routing, addressing, and packet forwarding
  - Data Link Layer (L2)- Manages node-to-node data transfer and error detection
  - Physical Layer (L1) - Handles the physical connection and raw data transmission
- Purpose
  - The OSI model serves as a universal set of guidelines for ensuring interoperability among diverse networking products and software
- Where did TCP/IP come from?
  - Developed in the 1970s for the United States Department of Defense or DoD's Advanced Research Projects Agency Network (ARPANET) project.
- What is TCP/IP?
  - Also known as the TCP/IP suite and Internet Protocol Suite
  - A conceptual framework used for designing and implementing network communications protocols in networks. It consists of four layers.
- Layers of the TCP/IP Model
  - Application Layer (L4) - focused on user interfaces and higher-level protocols.
  - Transport Layer (L3) - responsible for end-to-end communication and data transfer management.
  - Internet Layer (L2) - handles routing and addressing of data packets for transmission across networks.
  - Network Interface Layer (L1) - Ffocused on the network interface for data exchange over physical networks.

- Additional Resources:
  - If applicable

Purpose and Timeframe:

Introduced in the late 1970s, the model aimed to standardize computer network protocols and ensure interoperability among diverse network technologies.